

Bell X-1



“The four chambers blow a 30-foot [9-m] lick of flame. Christ, the impact nearly knocks you back into last week!”

CHUCK YEAGER

ON FLYING THE X-1 ON FULL THRUST

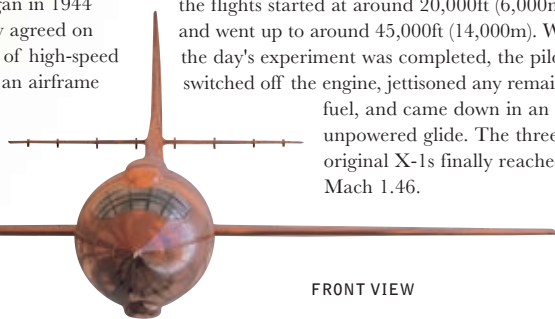
“ANYONE WITH BRAIN CELLS would have to wonder what in hell he was doing in such a situation – strapped inside a live bomb that’s about to be dropped out of a bomb bay.” That was Chuck Yeager’s description of sitting inside the Bell X-1 rocket plane, waiting for release from its B-29 mother ship. A “live bomb” was not a bad description for the X-1, which was designed as a manned projectile with wings. Intended for ground launch, the X-1 was air-launched to maximize its flying time – two and a half minutes on full power. It would have used up all its fuel climbing toward operational altitude.

The X-1 was a strictly experimental aircraft, equipped to collect data. Design began in 1944 when the USAAF and NACA jointly agreed on the need to investigate the problems of high-speed flight. The Bell design team created an airframe maximized both for speed and high strength – the chief concern was that the aircraft stand up to what were expected to be exceptional

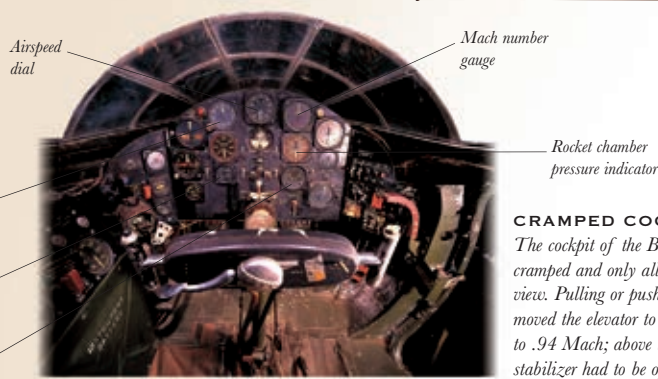
stresses or buffeting. As an extra safety feature Bell engineers made the stabilizer – the horizontal tailplane – movable so that it could be used for pitch control if the elevators failed.

The X-1 was fueled by alcohol and liquid oxygen (LOX), which had to be kept at -297°F (-182.7°C). The LOX was stored directly behind the pilot, making this, according to Yeager, “the coldest airplane ever flown.” Piloting the X-1 was “like trying to work and concentrate inside a frozen food locker.”

The pilot had the choice of igniting the four rocket chambers one at a time or in combination; all four together gave maximum thrust. Most of the flights started at around 20,000ft (6,000m) and went up to around 45,000ft (14,000m). When the day’s experiment was completed, the pilot switched off the engine, jettisoned any remaining fuel, and came down in an unpowered glide. The three original X-1s finally reached Mach 1.46.



FRONT VIEW



Airspeed dial

Mach number gauge

Rocket chamber pressure indicator

Altitude indicator

Stabilizer position

Fuel supply

CRAMPED COCKPIT

The cockpit of the Bell X-1 was extremely cramped and only allowed the pilot a limited view. Pulling or pushing the control wheel moved the elevator to give pitch control up to .94 Mach; above this speed the movable stabilizer had to be operated.

Specifications

ENGINE	Reaction Motors XLR11-RM3 rocket engine
WINGSPAN	28ft (8.5m)
LENGTH	30ft 10in (9.4m)
HEIGHT	13ft 5in (3.3m)
WEIGHT	12,250lb (5,557kg)
TOP SPEED	967mph (1,556kph)

Pressurized cockpit

Fuselage – and even the windshield – is shaped like a .50 caliber bullet

Horizontal stabiliser

GLAMOROUS GLENNIS

Chuck Yeager named his sound-barrier-breaking Bell X-1 Glamorous Glennis in tribute to his wife. The aircraft was painted bright orange because it was thought this would help cameras and chase aircraft tracking the flights. It was later found that white was a better color for the purpose.

Thin yet exceptionally strong wing sections improved flying control

